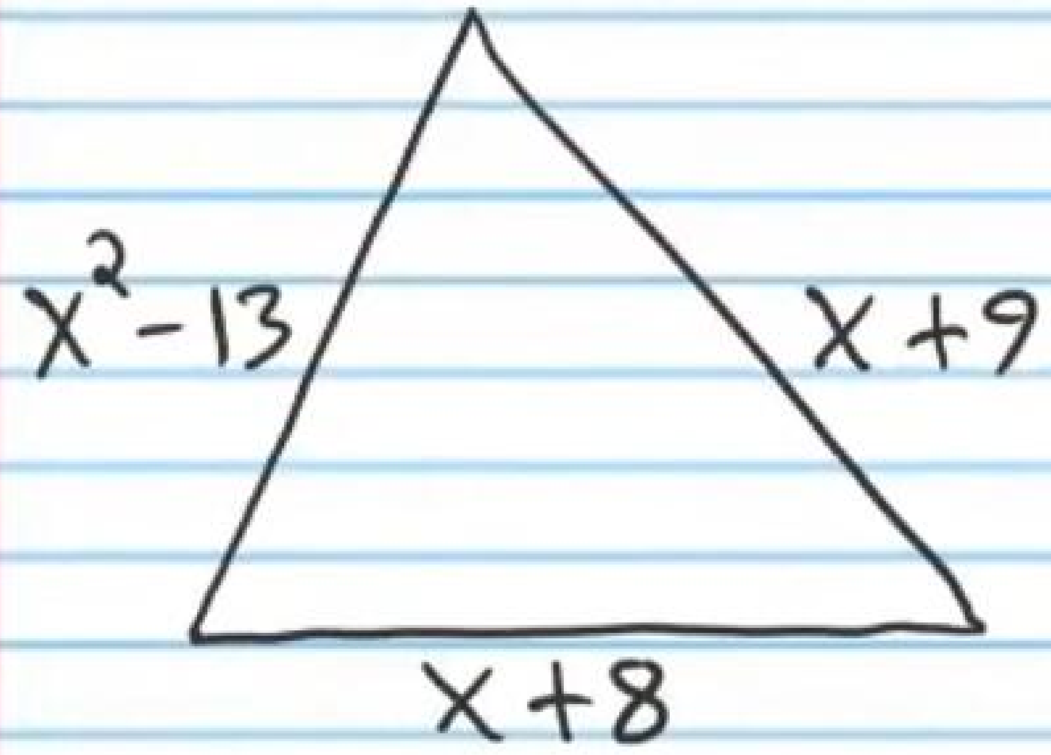
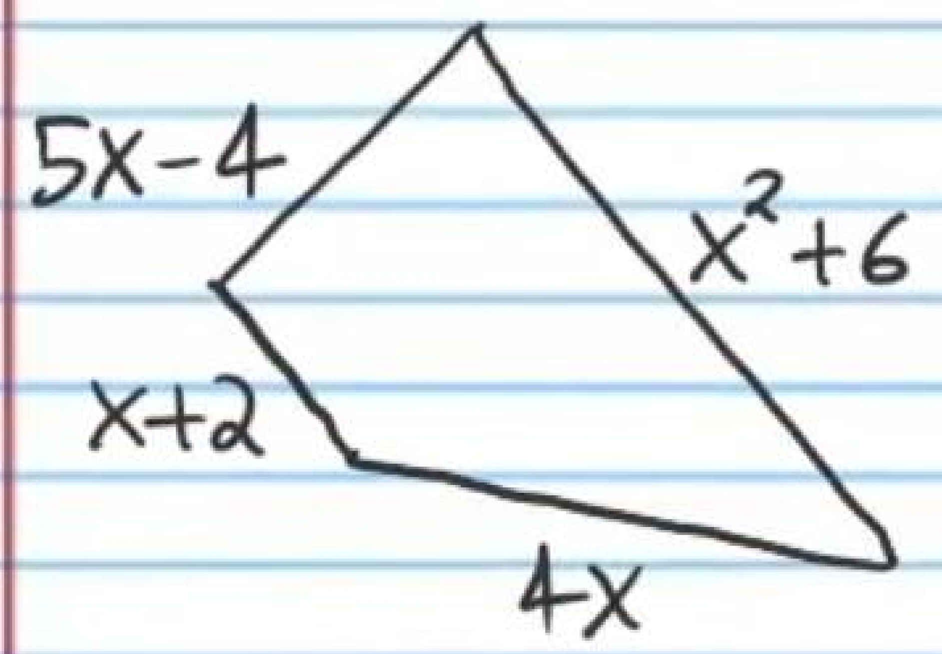
Name
Date
Course

Applications of Algebra (Part II) Homework

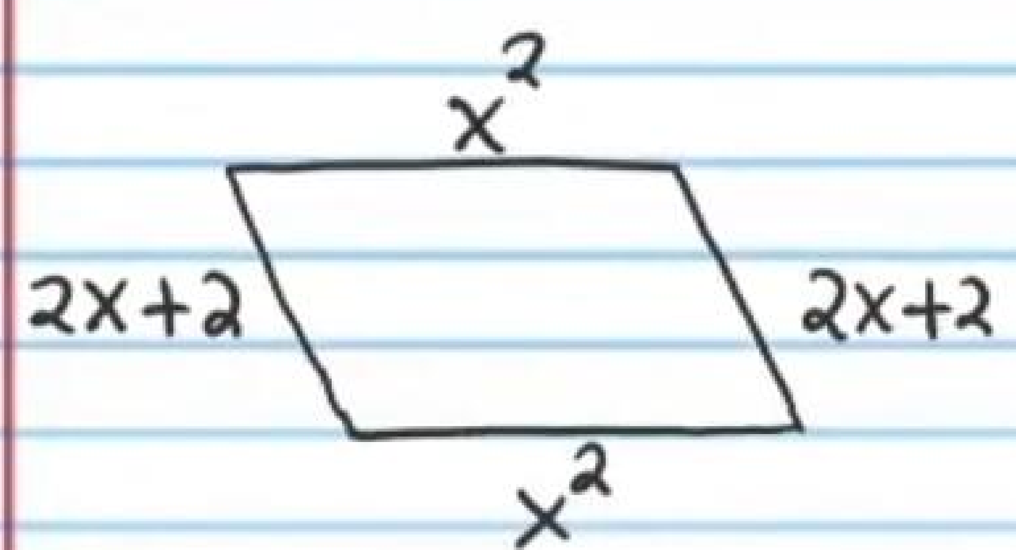
① The perimeter of the triangle below is 39 meters. Find the length of each side.



② The perimeter of the quadrilateral (i.e. a four-sided shape) below is 28 centimeters. Find the length of each side.



③ The perimeter of the parallelogram below is 34 inches. Find the length of each side.



Use the graphing method to solve the three systems below. One system below cannot be solved using the graphing method. Identify the system that cannot be solved by graphing and use the graphing method to explain why.

☆ You must use graphing paper.

④ $\begin{cases} y=x \\ x+y=4 \end{cases}$ ⑤ $\begin{cases} y+5x=-2 \\ -2x+1=y \end{cases}$ ⑥ $\begin{matrix} 2x-3y=-7 \\ 4x-2y=1 \end{matrix}$

⑦ A boy kicks a ball into the air. Neglecting air resistance, the height (in feet) of the ball after t seconds in the air can be found with the following equation.

$$h=-16t^2+48t$$

a) Find the height of the ball after two seconds in the air. b) How long will it take for the ball to reach a height of 20 feet (Hint: there are two answers)? c) How long will it take for the ball to hit the ground (Hint: there are two answers—one is relevant and the other is not)?

⑧ A woman standing on a bridge drops a coin into the creek below. The height of the coin (in feet) t seconds after free fall can be modeled by the following equation (the equation assumes no air resistance).

$$h=-16t^2+40$$

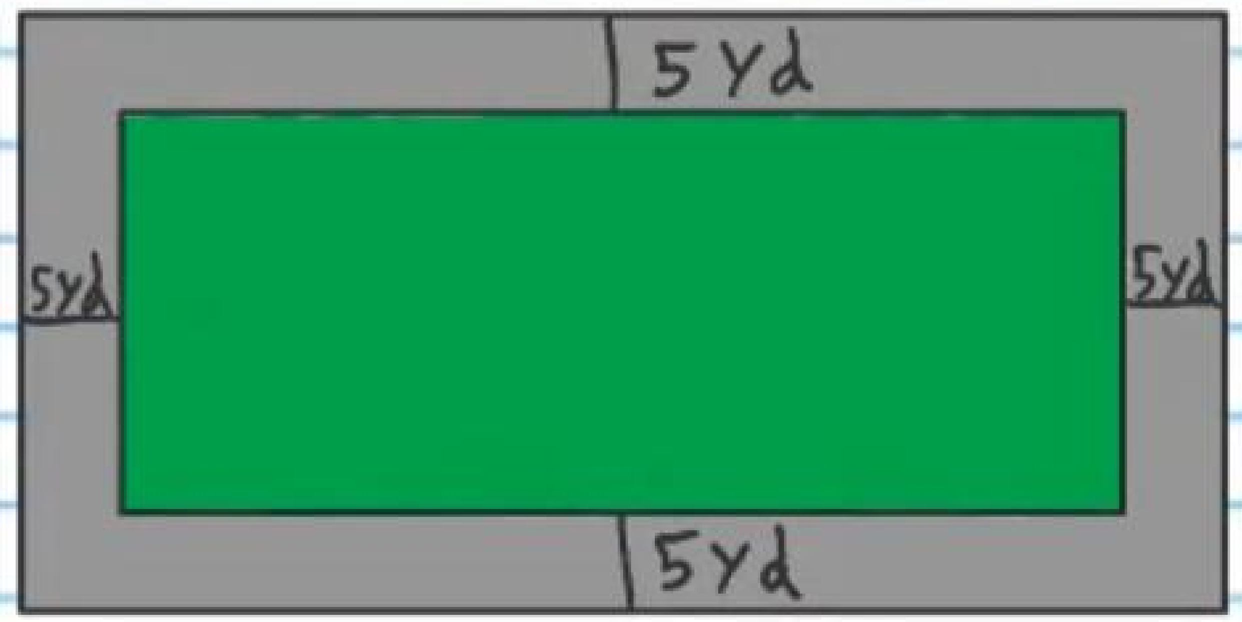
a) What is the height of the coin $2/3$ of a second after it is dropped? b) After how many seconds of free fall will the coin reach a height of 20 feet? c) When will the coin hit the ground?

⑨ A man throws an object from a window 90 feet above the ground. Neglecting air resistance, the height of the object (in feet) t seconds after it is thrown can be modeled by the equation below.

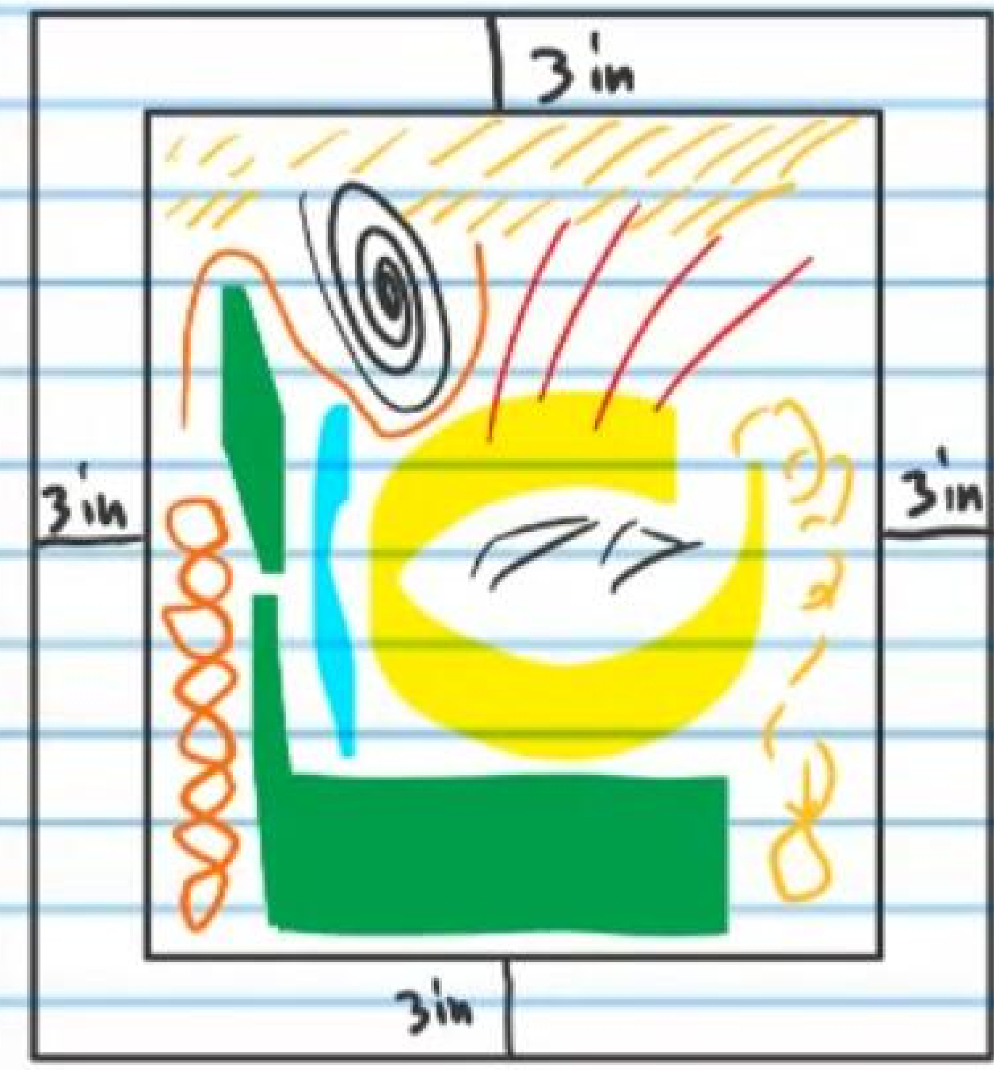
$$h = -16t^2 - 4t + 90$$

a) What is the height of the object one second after it is thrown? b) How long will it take for the object to descend to a height of 18 feet? c) How long will it take for the object to hit the ground?

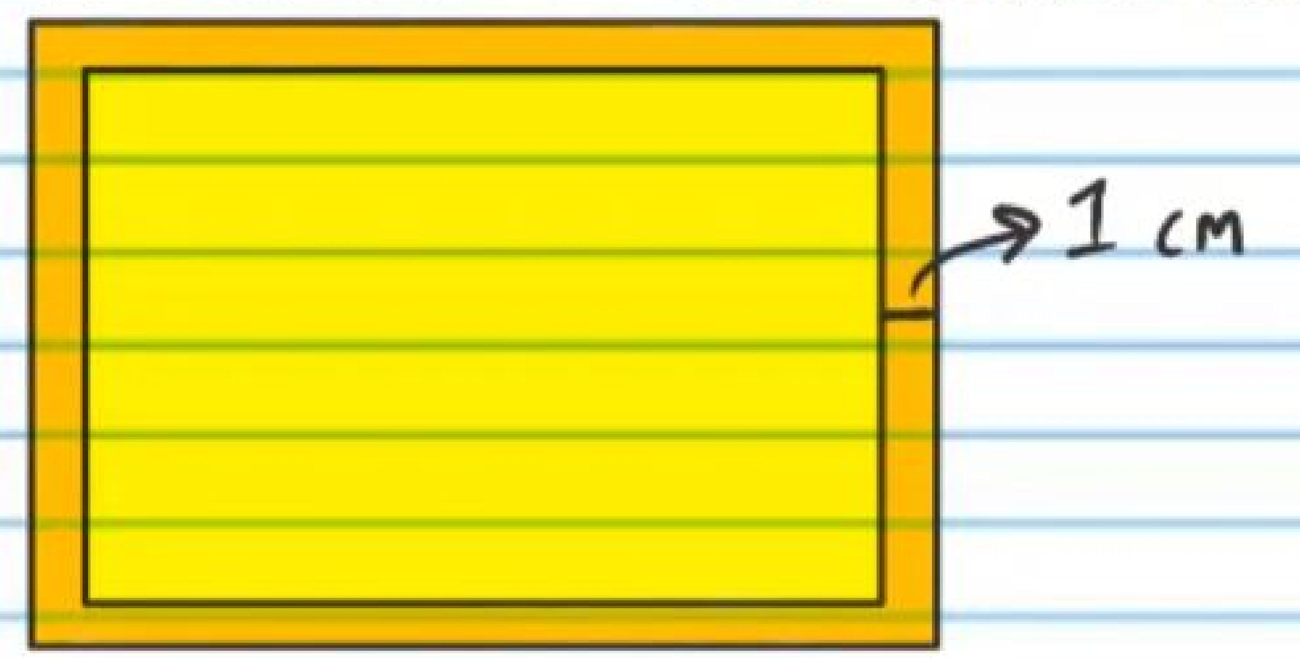
⑩ A road surrounds a rectangular park. The width of the park is 30 yards more than the length. The area of the road and the park combined is 800 square yards more than the area of the park alone. What are the dimensions of the park?



⑪ A painting with a flat white frame is illustrated below. The area of the painted portion and frame combined is 348 square inches more than the area of the painted portion. The width of the painted portion is 16 inches less than the length of the painted portion. What are the dimension of the whole painting (frame and painted portion combined)?



⑫ The area (cheese and crust) of a rectangular cheese pizza is 234 square centimeters more than the area of the cheese portion alone. The length of the cheese portion is 23 centimeters less than the width. What are the dimensions of the entire pizza? The crust is one centimeter thick.



Answers

1. $\boxed{12\text{ m}, 13\text{ m}, 14\text{ m}}$
2. $\boxed{4\text{ cm}, 6\text{ cm}, 8\text{ cm}, 10\text{ cm}}$
3. $\boxed{8\text{ in}, 8\text{ in}, 9\text{ in}, 9\text{ in}}$
4. $\boxed{x = 2}$, $\boxed{y = 2}$
5. $\boxed{x = -1}$, $\boxed{y = 3}$
6. $\boxed{x = 2\tfrac{1}{8}}$, $\boxed{y = 3\tfrac{3}{4}}$ This system cannot be solved with the graphing method because the point of intersection does not occur at a corner of a square, which makes it difficult to determine its exact location.
7. a) $\boxed{32\text{ ft}}$ b) $\boxed{\tfrac{1}{2}\text{ s and } 2\tfrac{1}{2}\text{ s}}$ c) $\boxed{3\text{ s}}$
8. a) $\boxed{32\tfrac{8}{9}\text{ ft}}$ b) $\boxed{\frac{\sqrt{5}}{2}\text{ s}}$ c) $\boxed{\frac{\sqrt{10}}{2}\text{ s}}$
9. a) $\boxed{70\text{ ft}}$ b) $\boxed{2\text{ s}}$ c) $\boxed{2\tfrac{1}{4}\text{ s}}$
10. $\boxed{20\text{ yd} \times 50\text{ yd}}$
11. $\boxed{40\text{ in} \times 24\text{ in}}$
12. $\boxed{71\text{ cm} \times 48\text{ cm}}$